Chapter 16: Continuum Mechanics

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There are three main studies of mechanics; there are the mechanics of point masses, mechanics of rigid bodies, and continuum mechanics. Other areas of mechanics, like classical mechanics, often times do not account for the variability of the mass and position of certain systems. Continuum mechanics studies how mass is distributed over certain regions in a system and how the relative positions of the various parts can change in arbitrary and continuous manner. For example, water flowing down a pipe would be studied with classical mechanics. Here, we explore one specific case of a wave on a taut string. We discover the usage of the general wave equation, and plot different graphs depicting this wave through time.

Classical Mechanics is the study of how things move. It encompasses everything from projectiles and charged particles, momentum, and oscillations. However, classical mechanics cannot account for all forms of mechanics. For example, when things approach the speed of light, you can no longer use classical mechanics. Simply put, classical mechanics excludes relativity and quantum mechanics. However, it has usages that remain extremely important to every day life.

There are three main divisions in mechanics. In order of increasing complexity, there is the mechanics of point masses, mechanics of rigid bodies, and continuum mechanics. The last piece acknowledges some of the simplifications that are made when calculating other mechanical problems. For example, continuum mechanics accounts for how mass is distributed over all regions in a system, and how the relative positions of the various parts can change in any arbitrary and continuous manner.

These complex system are comprised of examples like the flow of air past the wing of an airplane, the motion of a fluid, and water down a pipe. Because these systems are composed of an infinite number of parts, they also have an infinite number of coordinates, which can make solving problems for these systems very complicated.

One of the fascinating and prevalent problems to arise from continuum mechanics is the problem of waves. For a taut string, we can make some simplifying assumptions to get to a general solution for one-dimension, two-dimensional, and three-dimensional wave equations.

First we will need to understand more about taut strings before we can start to solve for different systems. For a taut string, the configuration of the string at any time can be given by $u(x$. In a discrete case with known individual masses of the string, the displacements u depend on time and become $u_i(t)$. Here, i is the index $i = 1...n$. In the continuous case, we write $u(x,t)$, where u depends on two variables. Now, as we will see later, this then causes Newton's second law to become a partial differential equation in order to solve for u.

In addition, there are a couple simplifying assumptions that we will make. First, we will assume that gravity is not greatly affecting our system and we will therefore ignore it. Next, we will use that all displacements $u(x,t)$ are so small, that the string is almost parallel to the x-axis. We do this so we can assume that tension is uniform throughout the string, and thus the same for all x and t values.

Now we can explore the motion of a taut string. Generally, the wave equation has three solutions which describes the different components of a wave. It can be a disturbance $u(x,t)$ that travels rigidly along a string from left to right, a disturbance $u(x,t)$ that travels rigidly along a string right to left, or any combination of the two.

First, let's contemplate a two-dimensional wave on a taut string. This wave is on a string with a length $L = 2$. At $t = 0$, it has initial position given by

$$u(x,0) = \begin{cases} 2x, & \text{if } 0 \leq x \leq 1/2 \\ 2(1-x) & \text{if } 1/2 \leq x \leq 1 \\ 0 & \text{if } x > 0 \end{cases} \tag{1}$$

Here, the wave also has an initial speed of $c = 0.5$ m/s. Now, we want to look at the initial displacement of the string at $t = 0$. To do this, we can use the help of a python program in Jupyter Notebooks. The code to create the graph is in Figure 1.

```
def fun (t,x):
    if 0 <= x <= 0.5:
        return float(2*x)
    elif 0.5 <= x <= 1:
        return 2*(1-x)
    return 0

vfun = np.vectorize(fun)

x = np.linspace(0, 2, 1000)
t = vfun(2, x)
```

FIG. 1: The code needed to create the initial graph for our wave using Equation 1.

Here, what we are doing is defining a function for our initial position of our string. Then, we give it the parameters set by our piece-wise function for $u(x,0)$. After we create this, we vectorize our values and tell it to complete this from 0 to 2. Finally, we simply graph this as seen in Figure 2.

Now, we can further our understanding of the motion of our wave through time. Normal modes determine all

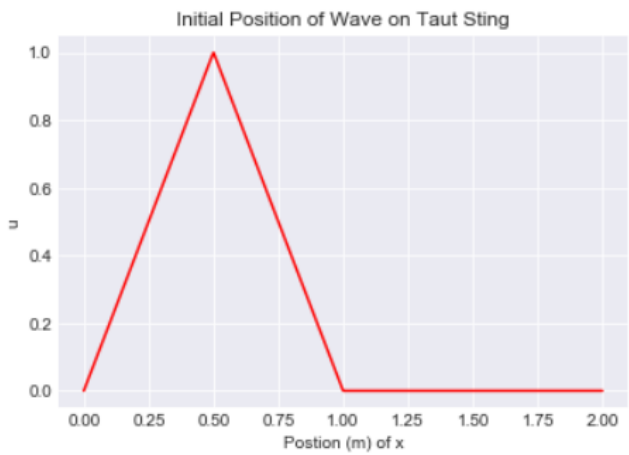

FIG. 2: The initial position of the wave on the taunt string of length $2m$ and with a speed of $c = 0.5\ m/s$.

possible motion of our finite string. Any possible motion can be expanded by the normal-mode solutions. We know that any motion of the string is given by $u(x,t)$, that satisfies the wave equation which can be seen in Equation 2.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial t^2} \tag{2}$$

Since we have a finite string, it must also satisfying the boundary conditions, given by Equation 3, that fix the position u and the velocity $\dot{u}$ for all times t.

$$u(0,t) = u(L,t) = 0 \tag{3}$$

The normal mode is also determined by its initial positions $u(x,0) = u_0(x)$ and its initial velocity $\dot{u}(x,0) = \dot{u}(x)$. With all of this taken into account, we can rewrite the normal mode solution in terms of "sine plus cosine" as seen in Equation 4.

$$u(x,t) = \sin k_n x (B_n \cos w_n t + C_n \sin w_n t) \tag{4}$$

The claim here, is that we can use this to find any motion of this finite wave as a linear combination, which can be seen in Equation 5.

$$u(x,t) = \sum_{n=1}^{\infty} \sin k_n x (B_n \cos w_n t + C_n \sin w_n t) \tag{5}$$

Here, at $t = 0$ our velocity is zero. Thus in our initial equation, we can omit the right side of the equation that depends on the constant C_n. Therefore, our equation becomes a Fourier time sine series, as shown in Equation 6, where the coefficient "B_n can be chosen so $u(x,0)$ matches any given initial value $u_0(x)$[3]" (Taylor, 692).

$$u(x,0) = \sum_{n=1}^{\infty} B_n \sin k_n x \tag{6}$$

Using this, we can determine the Fourier sine coefficients for any value of B_n. I initially found the first term, B_1, by hand and then created a program in Jupyter Notebook to find the next 19 terms. Since our string is initially at rest and we do not have to account for C_n, our equation for B_n becomes

$$B_n = \frac{2}{L} \int_0^L u_0(x) \sin \frac{n\pi x}{L} dx \tag{7}$$

Given our initial constraints in Equation 1, we can substitute in our known variables and the bounds of our integrals. This gives us the equation

$$B_1 = 2 \int_0^{\frac{1}{2}} x \sin \frac{\pi x}{2} dx + 2 \int_{\frac{1}{2}}^{1} (1-x) \sin \frac{\pi x}{2} dx \tag{8}$$

Now, we can solve these integrals by using u-substitution and integration by parts. To do this, we can solve our two integrals separately and then add their sum together at the end. Our first integral, which we can call r, is solved for below

$r = 2 \int_0^{\frac{1}{2}} x \sin \frac{\pi x}{2} dx$

$r = 2 \int_0^{\frac{1}{2}} \frac{4u \sin u}{\pi^2} du$

$r = 2(\frac{4}{\pi^2}) \int_0^{\frac{1}{2}} u \sin u du$

$r = 0.123$

Then, we can solve for our second integral, which we will call a.

$a = 2 \int_{\frac{1}{2}}^{1} (1-x) \sin \frac{\pi x}{2} dx$

$a = 2 \int_{\frac{1}{2}}^{1} \sin \frac{\pi x}{2} dx - \int_{\frac{1}{2}}^{1} x \sin \frac{\pi x}{2} dx$

Through u-substitution and much simplification, we get

$a = 2(\frac{1.41}{\pi} - 0.344) = 0.2127$

Now we have the solution, $r + a = 0.123 + 0.2127 = 0.3357 = B_1$. To avoid having to do this every time by hand, utilizing online programming made calculating the next 19 terms easy. Creating a for loop, we can have the program calculate each value of B for n ranging from 1 to 20. Then, we could put those values into a useful table, as shown in Table I

Now, with all of these values of B_n, we can make plots of the solutions. To do this, I also utilized python coding, as seen in Figure 3.

First, we create an array for x that starts at 0 and is supposed to stop at 2 with 1000 points between them. Then, we set our initial graph to display at $t = 0$. From

B_n	Value
B_1	0.3357
B_2	0.405
B_3	0.7245
B_4	0.6366
B_5	0.4347
B_6	0.2122
B_7	0.053274
B_8	0
B_9	0.0414
B_{10}	0.1273
B_{11}	0.1976
B_{12}	0.2122
B_{13}	0.1672
B_{14}	0.0909
B_{15}	0.0249
B_{16}	0
B_{17}	0.0219
B_{18}	0.070
B_{19}	0.1144
B_{20}	0.1273

TABLE I: All of the values of B_n for n from 1 to 20.

```
x = np.linspace(0,2,1000)
t = 0 # initial time
u = 0 # start u at 0
for n in range(20):
    u = u + np.sin(ks[n]*x) * Bs[n] * np.cos(ws[n]*t)  # equation 16.30
plt.plot(u)
plt.show()
```

FIG. 3: Code depicting the loop needed to graph $u(x,t)$ at different periods of time.

this point, we also need to provide a value of $u = 0$, which we will add onto in our for loop. In our loop, we take all integers of n from 0 to 20 and calculate them with our known equation for u. Finally, we simply plot the graph.

Our first graph is depicted at $t = 1$ in Figure 4. I then created graphs at different time steps, including the one shown in Figure 5, which is at $t = 3$ s.

I then created an animation of the string through time. Initially, one might be quite puzzled by the difference between Figure 2 and Figure 4. However, with a little analysis, we can understand some of the meaning behind each of the graphs. The first graph depicts the initial displacement of the string. While this approximation is fairly strong, it fails to produce a sharp turn at the apex. This can be easily understood, since no finite sum of sines and cosines can produce an instantaneous change of slope. Then, in our next graphs, I believe we are seeing the different values of B_n and the position of the wave under some of the affects of the boundaries. Since the ends of our strings are tied down and not free to move, the ends can create some force on the rest of the string,

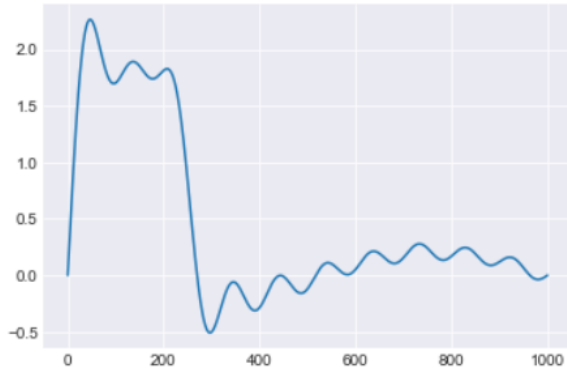

FIG. 4: Graph depicting the position of the wave at $t = 1$.

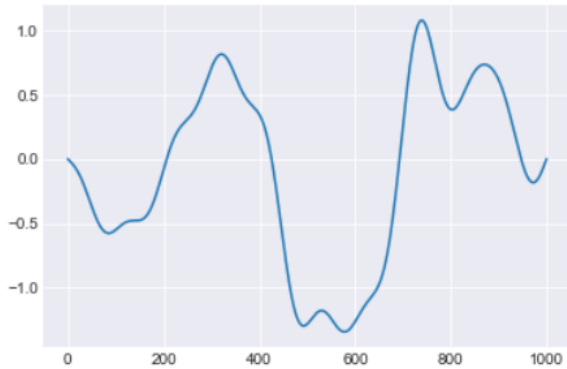

FIG. 5: Graph depicting the position of the wave at $t = 3$.

influencing it motion. The graphs I created at different time steps are much more sporadic in nature than I was anticipating. It would have made more sense to see a sinusoidal wave. This leads me to believe that there could be a possible error in the graph.

As the wave approaches the right end of the string, the wave seems to almost reflect across the x-axis before moving to the left. Then at $t = 8$, we get a graph identical to $t = 1$. This pattern continues, where the wave moves along the x-axis in the same pattern for the first $20s$.

While my data are not perfect, they depict the wave on the taunt string through time and display the different values of B_n for the first 20 terms. While some of the calculus was straight forward, creating some of the loops in Jupyter notebook proved to be a challenge. Keeping all of the values in organized lists and having a clear understanding of what the loop needed to accomplish was key to any success in this project.

Continuum mechanics is a fascinating subject. While classical mechanics addresses important concepts like tracking point particles, discovering the complex nature of intricate systems is very important to understand different environments.

Taylor, John R. Classical Mechanics. University Sci-

ence Books, 2005.